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Project Proposal

CSE331L: Microprocessor Interfacing & Embedded Systems Lab

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Project Title:

Smart Fire, Gas, and Environment Safety Monitoring System Using STM32F103C8T6

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Smart Fire, Gas, and Environment Safety Monitoring System

Using STM32F103C8T6

A Multi-Sensor Embedded Safety Monitoring Prototype

1. Abstract

Fire hazards and unsafe environmental conditions in rooms, laboratories, and workshops often develop quickly and are missed by single-purpose detectors that monitor only one parameter at a time. This project proposes an STM32F103C8T6-based embedded system that monitors four different safety-related parameters together: temperature and humidity (DHT11), object/human presence through distance (HC-SR04), direct flame presence (flame sensor), and smoke and flammable gas concentration (MQ-2). The STM32F103C8T6 reads all four sensors through its GPIO, ADC, and timer peripherals, classifies each reading against a predefined threshold, and then applies a combined decision rule to classify the overall environment as SAFE, WARNING, or CRITICAL. The result is shown continuously on a 0.96-inch I2C OLED, while a buzzer and a red/yellow/green LED set give an instant audible and visual warning whenever an unsafe condition is detected. The project is expected to produce a low-cost, functional prototype that demonstrates a complete input-processing-output embedded pipeline suitable for basic indoor fire and environmental safety monitoring.

2. Introduction and Motivation

2.1 Introduction

Fires and unsafe environmental changes can begin quietly - a small gas leak, a rising room temperature, or a spark - well before they become visible or obviously dangerous. Many low-cost commercial detectors respond to only one such sign, for example smoke alone, and therefore miss situations where two or three smaller warning signs occur together. An embedded system that reads several sensors at once and reasons about them jointly can catch this kind of situation earlier and more reliably.

This project builds such a system around the STM32F103C8T6 (Blue Pill), a low-cost ARM Cortex-M3 microcontroller commonly used in student and hobbyist embedded projects because of its sufficient GPIO count, built-in ADC channels, timers, and I2C support. Several published STM32-based safety-monitoring designs combine similar sensors - temperature/humidity, flame, gas, and an OLED with buzzer alerts - which supports both the technical feasibility and practical value of this approach.

2.2 Motivation

The motivation for this project is to build a working, affordable safety-monitoring prototype using parts that are already common in an undergraduate embedded-systems lab, while still covering a meaningfully wide range of hazards. In particular, the system aims to:

- Detect abnormal temperature and humidity levels.
- Detect smoke and common flammable gases.
- Detect a visible flame directly, independent of the gas/smoke reading.
- Detect whether a person or object is present within a defined zone.
- Combine all of the above into one real-time decision instead of multiple separate alarms.
- Give an immediate, easy-to-understand visual and audible warning.

Such a system is directly useful in a small classroom, laboratory, workshop, or server/equipment room, where several basic safety parameters need to be watched at the same time with a single, low-cost device.

3. Objectives

The project has the following measurable objectives:

- Build an STM32F103C8T6-based data-acquisition system that reads at least four sensing inputs in real time.
- Measure temperature and humidity with the DHT11 and display the readings on an OLED.
- Detect flammable gas/smoke and flame using the MQ-2 and flame sensor, raising an alert once a defined threshold is crossed.
- Detect presence/proximity with the HC-SR04.
- Implement threshold-based decision logic that fuses all four readings into a single SAFE / WARNING / CRITICAL status shown through the OLED, buzzer, and LEDs.

4. System Overview

4.1 Overall Concept

The system has three stages: sensing, processing, and output. In the sensing stage, four modules continuously feed data to the STM32F103C8T6. In the processing stage, the microcontroller samples each sensor, classifies the reading against a threshold, and combines the classifications using the decision logic described in Section 7. In the output stage, the current status and live values are shown on the OLED while the buzzer and LEDs give an immediate physical warning when the status is WARNING or CRITICAL.

4.2 Block Diagram

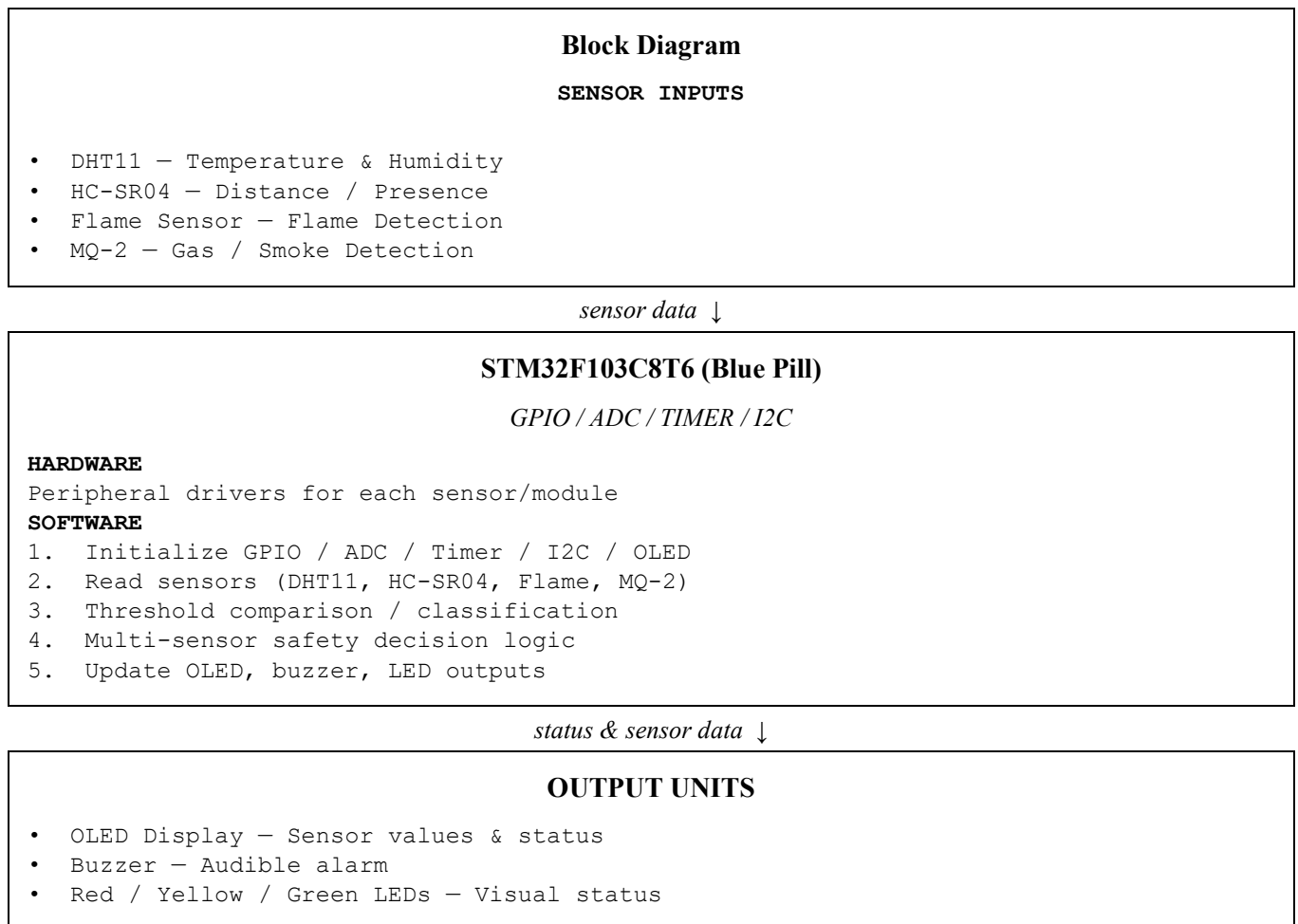


Figure 1: System block diagram - sensing, processing (hardware + software), and output stages

4.3 Module Functions

- **DHT11:** reads temperature and relative humidity so the system can judge whether the room is within a comfortable/safe range.
- **HC-SR04:** measures distance using an ultrasonic pulse-echo method; the measured distance is converted into a simple presence/proximity decision for a defined monitoring zone.
- **Flame Sensor:** gives a direct, independent flame reading that feeds straight into the critical-alert logic.
- **MQ-2:** provides an analog gas/smoke concentration reading through the ADC, used to detect smoke and common flammable gases such as LPG and methane.
- **OLED (0.96-inch, I2C):** shows live temperature, humidity, gas/smoke level, flame status, distance, and the overall safety status.
- **Buzzer and LEDs:** the buzzer sounds during WARNING/CRITICAL states; the green, yellow, and red LEDs give an instant visual read of Safe, Warning, and Critical states respectively.

5. Hardware Components

The table below lists every hardware component required for the project, assuming none of the parts are currently available and all must be purchased, along with a short justification for each and its approximate cost.

Component	Qty.	Justification	Approx. Cost (Tk.)
STM32F103C8T6 (Blue Pill)	1	Central microcontroller; provides GPIO, ADC, timers and I2C needed by all sensors and the display	250 - 350
ST-LINK V2 Programmer	1	Used to flash and debug the STM32	300 - 450
DHT11 Sensor	1	Measures temperature and humidity for environmental monitoring	120 - 150
HC-SR04 Ultrasonic Sensor	1	Measures distance for presence/proximity detection	90 - 120
Flame Sensor Module	1	Detects visible flame directly for fire alerting	60 - 70
MQ-2 Gas/Smoke Sensor	1	Detects smoke and flammable gases (LPG, methane, etc.)	140 - 160
0.96" I2C OLED Display	1	Displays live sensor readings and system safety status	250 - 350
Active Buzzer	1	Generates an audible alarm during warning/critical states	30 - 50
Red / Yellow / Green LED	3	Give a quick visual indication of Safe / Warning / Critical state	20 - 40
220 ohm Resistor	3	Current-limiting resistors for the three LEDs	10 - 20
Breadboard (830-point)	1	Prototyping platform for wiring all modules together	~140
Jumper Wires	As required	Connects sensors, display and MCU on the breadboard	~185

Table 1: Hardware components, justification and cost breakdown

Estimated Total Project Cost

Component	Cost
STM32F103C8T6 (Blue Pill)	Tk 250 - 350
ST-LINK V2 Programmer	Tk 300 - 450
DHT11 Sensor	Tk 120 - 150
HC-SR04 Ultrasonic Sensor	Tk 90 - 120
Flame Sensor Module	Tk 60 - 70
MQ-2 Gas/Smoke Sensor	Tk 140 - 160
0.96" I2C OLED Display	Tk 250 - 350
Active Buzzer	Tk 30 - 50
LEDs (3 pcs)	Tk 20 - 40
220 ohm Resistors (3)	Tk 10 - 20
Breadboard	Tk ~140
Jumper Wires	Tk ~185
Approximate Total	Tk 1595 - 2085

Note: actual cost may vary by shop and component quality.

6. Software Design

6.1 Programming Language and Development Tools

- Programming Language: C
- Microcontroller: STM32F103C8T6
- IDE: Keil uVision / STM32 development environment
- Programmer / Debugger: ST-LINK V2

6.2 Communication Protocols and Peripheral Interfaces

Each device uses a different STM32 peripheral or communication method, summarized below.

Device	STM32 Interface / Method
DHT11	Single-wire, timing-based GPIO communication
HC-SR04	GPIO trigger pulse + timer-based echo measurement
Flame Sensor	Digital GPIO input (or ADC for analog output variants)
MQ-2	Analog input via ADC (digital GPIO for threshold output)
OLED	I2C
Buzzer	GPIO output
LEDs	GPIO output

Table 2: Sensor-to-peripheral interface mapping

6.3 Algorithms and Control Logic

Threshold-Based Classification

```
Temperature : <20C Cold | 20-30C Comfortable | 30-35C Warm | >35C Hot
Humidity    : <40% Dry | 40-70% Comfortable | >70% Humid
Distance    : below threshold -> Presence Detected; else -> Area Clear
Gas/Smoke   : compared with calibrated threshold -> Safe / Warning / Danger
Flame       : sensor triggered -> Alert; not triggered -> Safe
```

Figure 2: Threshold classification applied to each sensor reading

Multi-sensor safety decision logic: instead of relying on any single sensor, the system combines all classified readings into one overall decision, as summarized below.

Multi-Sensor Safety Decision Logic

```
Normal State:      Temp normal, Gas safe, Flame not detected,
                   Presence normal => SYSTEM SAFE
Warning State:     High temp OR high gas/smoke
                   => WARNING
Fire Alert:        Flame detected OR (high gas/smoke AND high temp)
                   => FIRE WARNING (buzzer + red LED)
Critical Fire Alert: Flame detected AND (high temp OR high gas/smoke)
                   AND/OR presence detected
                   => CRITICAL ALERT (buzzer + red LED + OLED screen)
```

Figure 3: Combined decision logic (Normal / Warning / Fire Alert / Critical Fire Alert)

Main Program Flow

```
START
-> Initialize GPIO, ADC, Timer, I2C, OLED
-> Read DHT11, HC-SR04, Flame Sensor, MQ-2
-> Process readings and compare with thresholds
-> Determine status: SAFE / WARNING / CRITICAL
-> Update OLED, LEDs and Buzzer accordingly
-> Repeat continuously
```

Figure 4: Main program execution flow

Pseudocode

```
INITIALIZE peripherals (GPIO, ADC, TIMER, I2C, OLED)
LOOP:
    read temperature, humidity from DHT11
    read distance from HC-SR04
    read flame_status from Flame Sensor
    read gas_level from MQ-2 (ADC)
    classify each reading against its threshold
    IF flame_status == DETECTED:
        status = CRITICAL ; buzzer = ON ; red_LED = ON
    ELSE IF gas_level > danger_threshold AND temperature > high_threshold:
        status = CRITICAL ; buzzer = ON ; red_LED = ON
    ELSE IF temperature > high_threshold OR gas_level > warning_threshold:
        status = WARNING ; buzzer = ON ; yellow_LED = ON
    ELSE:
        status = SAFE ; buzzer = OFF ; green_LED = ON
    update OLED with sensor values and status
    wait for next sampling interval
END LOOP
```

Figure 5: Control-logic pseudocode

6.4 Additional Notes on Communication

The OLED communicates over I2C, while the HC-SR04 and DHT11 use timer-based GPIO pulse timing. The MQ-2 is read through the STM32's internal ADC channel. No external wireless link is required for the core functionality; a UART connection through the ST-LINK's virtual COM port may optionally be used during development to monitor raw sensor values on a PC.

7. Expected Outcomes

Upon successful implementation, the system is expected to:

- Acquire real-time data from all four sensors using the STM32F103C8T6.
- Measure and display temperature and humidity on the OLED using the DHT11.
- Provide real-time distance measurement and presence/proximity detection using the HC-SR04.
- Detect flame directly and trigger a fire warning through the flame sensor.
- Detect elevated smoke/gas levels early using the MQ-2.
- Combine all sensor inputs into a single SAFE / WARNING / CRITICAL system status.
- Display live sensor values and the current safety status continuously on the OLED.
- Trigger the buzzer and the appropriate LED immediately during abnormal conditions.
- Demonstrate a complete Input - Processing - Output embedded-system pipeline using GPIO, ADC, timers, and I2C.
- Deliver a low-cost, functional prototype suitable for basic indoor fire and environmental safety monitoring.